ORCHID INTERNATIONAL COLLEGE

Tribhuvan University

Faculties of Computer Science

Bachelors of Science in Computer Science and Information Technology

(BSc. CSIT)

Course: Numerical Method (CSC-207)
Semester: III

A Lab Report On:

Numerical Differentiation & Integration

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LAB 3

TITLE: Numerical Differentiation & Integration

OBJECTIVES :

- To study Differentiation & Integration methods used to find the value using provided reference data.

THEORY:

1. Derivatives using forward difference:

Concept:

- The concept of derivatives using the forward difference method is a numerical approach to estimate the derivative of a function at a particular point by calculating the slope of the secant line passing through that point and a neighboring point slightly ahead of it. This method is based on the definition of the derivative, which is the rate of change of a function with respect to its independent variable.
- The forward difference approximation of the derivative is defined as:

$$f'(x) \approx \frac{f(x+h)-f(x)}{h}$$

Where:

- $f'(x)$ is the derivative of the function $f(x)$ at the point x.
- $f(x+h)$ is the value of the function at a point slightly ahead of x (using a step size h).
- $f(x)$ is the value of the function at point x.
- h is the step size used to determine the spacing between the two points.

Formula:

Newton forward interpolation formula is,

$$y(x) = y_0 + s\Delta y_0 + \frac{s(s-1)}{2!}\Delta^2 y_0 + \frac{s(s-1)(s-2)}{3!}\Delta^3 y_0 + \cdots \ldots \ldots \ldots ..$$

Where, $s = \frac{x-x_0}{h} \Longrightarrow x = x_0 + sh$

$$y(x_0 + sh) = y_0 + s\Delta y_0 + \frac{s^2-s}{2!}\Delta^2 y_0 + \frac{s^3-3s^2+2s}{3!}\Delta^3 y_0 + \cdots \ldots \ldots \ldots ..$$

Differentiating w.r.to. 's' we get,

$$y'(x) = \frac{dy}{dx}h = \Delta y_0 + \frac{(2s-1)}{2!}\Delta^2 y_0 + \frac{3s^2-6s+2}{3!}\Delta^3 y_0 + \cdots \ldots \ldots ....$$

$$\boxed{\therefore \frac{dy}{dx} = y'(x) = \frac{1}{h}\left[\Delta y_0 + \frac{(2s-1)}{2!}\Delta^2 y_0 + \frac{3s^2-6s+2}{3!}\Delta^3 y_0 + \cdots \ldots \ldots\right]}$$

Differentiating Again, we get,

$$\boxed{\frac{d^2y}{dx^2} = y''(x) = \frac{1}{h^2}[\Delta^2 y_0 + (s-1)\Delta^3 y_0 + \cdots \ldots \ldots]}$$

2. Integration using Simpsons 1/3 rule:

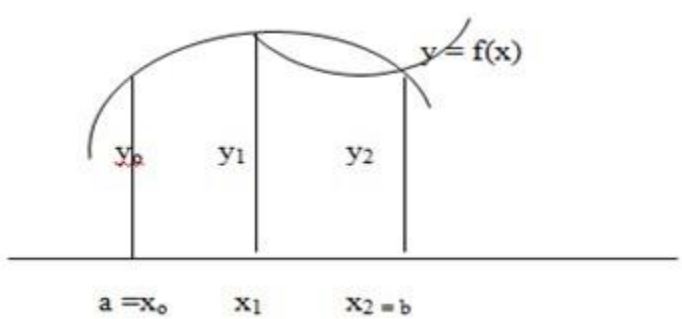
Let, function f(x) pow(sin(x),0.5)

Concept:

⇨

Here, the function $f(x)$ is approximated by a second-order polynomial $p_2(x)$ which passes through three sampling points as shown in fig. The three points include the end points $x_0(= a)$ and $x_2(= b)$ and a midpoint between them i.e. $x_0 = a$ and $x_2 = b$ and $x_1 = (a+b)/2$. The width of the segments h is given by

$$h = \frac{b-a}{2}$$



As we know from Newton's cotes formula,

$$\int_{x_0}^{x_n} f(x)dx = nh\left[f(x_0) + \frac{n}{2}\Delta f(x_0) + \frac{1}{12}(2n^2 - 3n)\Delta^2 f(x_0) + \frac{1}{24}(n^3 - 4n^2 + 4n)\Delta^3 f(x_0) + \cdots \ldots\ldots\right] \quad \ldots\ldots\ldots\ldots\ldots\ldots \text{(i)}$$

By putting $n = 2$ in eq. (i) and neglecting higher term we get,

$$\int_{x_0}^{x_2} f(x)dx = 2h\left[f(x_0) + \Delta f(x_0) + \frac{1}{6}\Delta^2 f(x_0)\right]$$

$$= 2h\left[f(x_0) + [f(x_1) - f(x_0)] + \frac{1}{6}[\Delta f(x_1) - \Delta f(x_0)]\right]$$

$$= 2h\left[f(x_1) + \frac{1}{6}[\{f(x_2) - f(x_1)\} - \{f(x_1) - f(x_0)\}]\right]$$

$$= 2h\left[f(x_1) + \frac{1}{6}[f(x_2) - 2f(x_1) + f(x_0)]\right]$$

$$= 2h\left[f(x_1) + \frac{1}{6}f(x_2) - \frac{1}{3}f(x_1) + \frac{1}{6}f(x_0)\right]$$

$$= 2h\left[\frac{1}{6}f(x_0) + \frac{2}{3}f(x_1) + \frac{1}{6}f(x_2)\right]$$

$$= \frac{2h}{6}[f(x_0) + 4f(x_1) + f(x_2)]$$

$$= \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)]$$

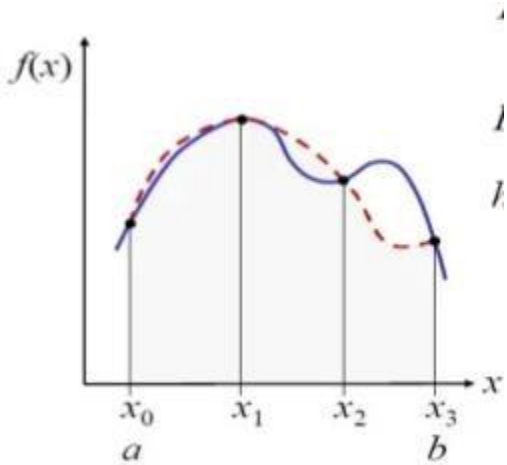
$$\boxed{\int_{x_0}^{x_2} f(x)dx = \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)]}$$

This equation is called *Simpson's 1/3 rule.*

3. Integration using Simpsons 3/8 rule:

Concept:

- ⇨ Simpson's 3/8 rule, also known as Simpson's Three-Eighths Rule, is a numerical integration technique used to approximate the definite integral of a function over a specified interval. It's an extension of the simpler Simpson's 1/3 rule and provides a way to estimate integrals when the number of subintervals is not divisible by 2 (i.e., when the number of intervals is not even). This rule offers higher accuracy than the 1/3 rule for a wider range of functions.



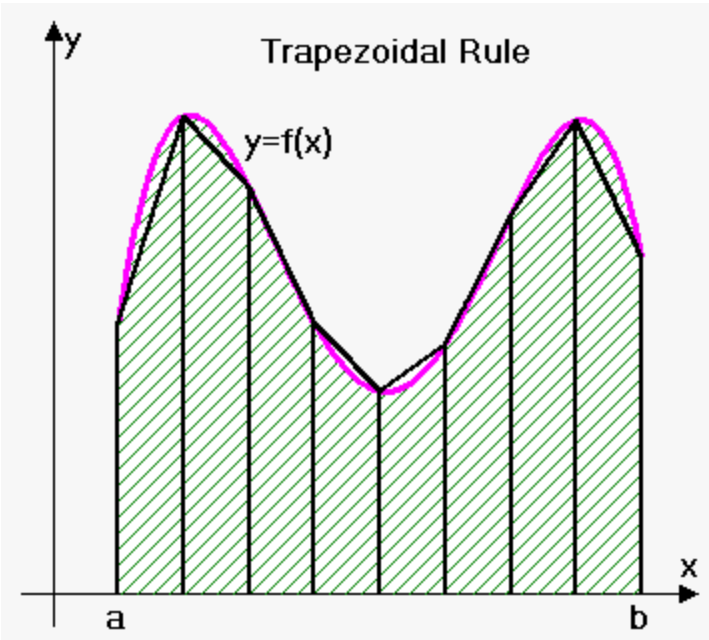
$$L(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} f(x_0)$$
$$+ \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} f(x_1)$$
$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} f(x_2)$$
$$+ \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} f(x_3)$$

$$\int_a^b f(x)dx \approx \int_a^b L(x)dx \;\; ; \quad h = \frac{b-a}{3}$$
$$= \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\right]$$

4. Integration using Trapeziodal rule :

Concept:

⇨ Let , f(x) be the function X0 = a & X1 = b.



h = X1– X0

As we know from Newton's cotes formula,

$$\int_{x_0}^{x_n} f(x)dx = nh\left[f(x_0) + \frac{n}{2}\Delta f(x_0) + \frac{1}{12}(2n^2 - 3n)\Delta^2 f(x_0) + \frac{1}{24}(n^3 - 4n^2 + 4n)\Delta^3 f(x_0) + \cdots \ldots\ldots\right] \quad \ldots\ldots\ldots\ldots\ldots\ldots \text{(i)}$$

By putting $n = 1$ in eq. (i) and neglecting higher term we get,

$$\int_{x_0}^{x_1} f(x)dx = h\left[f(x_0) + \frac{1}{2}\Delta f(x_0)\right]$$

$$= h\left[f(x_0) + \frac{1}{2}[f(x_1) - f(x_0)]\right]$$

$$= h\left[f(x_0) + \frac{1}{2}f(x_1) - \frac{1}{2}f(x_0)\right]$$

$$= \frac{h}{2}[f(x_0) + f(x_1)]$$

$$\boxed{\therefore \int_{x_0}^{x_1} f(x)dx = \frac{h}{2}[f(x_0) + f(x_1)]}$$

This equation is called *trapezoidal rule.*

PRACTICAL IMPLEMENTATION

TASK 1: Derivatives using forward difference

ALGORITHM :

1. Start
2. Read number of data (n)
3. Read data points for x and y:
   For i = 0 to n-1
      Read X_i and $Y_{i,0}$
   Next i
4. Read calculation point where derivative is required (xp)
5. Set variable flag to 0
6. Check whether given point is valid data point or not.
   If it is valid point then get its position at variable index
   For i = 0 to n-1
      If $|xp - X_i| < 0.0001$
         index = i
            flag = 1
         break from loop
      End If

   Next i
7. If given calculation point (xp) is not in
   x-data then terminate the process.
   If flag = 0
   Print "Invalid Calculation Point"
      Exit
   End If
8. Generate forward difference table
   For i = 1 to n-1
      For j = 0 to n-1-i
         $Y_{j,i} = Y_{j+1,i-1} - Y_{j,i-1}$
      Next j
      Next i
9. Calculate finite difference: $h = X_1 - X_0$
10. Set sum = 0 and sign = 1

11. Calculate sum of different terms in formula
    to find derivatives using Newton's forward
    difference formula:
    For i = 1 to n-1-index
        term = $(Y_{index, i})^i$ / i
        sum = sum + sign * term
        sign = -sign
    Next i
12. Divide sum by finite difference (h) to get result
    first_derivative = sum/h
13. Display value of first_derivative
14. Stop

OUTPUT :-

```
B:\3rd Sem\Lab\NM\Lab-3\Derivatives using forward difference.exe
Enter number of data: 6
Enter data:
x[0] = 1
y[0] = 7.989
x[1] = 1.1
y[1] = 8.403
x[2] = 1.2
y[2] = 8.781
x[3] = 1.3
y[3] = 9.129
x[4] = 1.4
y[4] = 9.451
x[5] = 1.5
y[5] = 9.750
Enter at what value of x you want to calculate derivative: 1.1
First derivative at x = 1.10 is 3.945852
                This program written by Aaryan Kushawaha ( 28900 )
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```

TASK 2: Integration using Simpsons 1/3 rule

Algorithm for Simpson's 1/3 rule

1. *Start*
2. *Read the value of lower limit of integration (a), upper limit of integration (b) and number of segments (n).*
3. *Compute* $h = (b - a)/n$
4. *for i=0 to n*
   $x_i = a + i * h$
   $y_i = f(x_i)$
   Repeat i
5. *Initialize so=se=0*
6. *for i=1 to n*
   if (i%2==1)
   so=so+y_i
   else
   se=se+y_i
   Repeat i
7. *ans=(h/3)*(* $y_1 + y_n + 4 * so + 2 * se$*)*
8. *Display ans as integral value.*
9. *END*

Output :-

B:\3rd Sem\Lab\NM\Lab-3\Integration using Simpsons 1_3 rule.exe

```
Function is f(x) pow(sin(x),0.5)
 Enter lower limit of integration: 0
Enter upper limit of integration: 30
Enter number of sub intervals: 15

Required value of integration is: -1.#IO
                This program written by Aaryan Kushawaha ( 28900 )
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TASK 3: Integration using Simpsons 3/8 rule

ALGORITHM:

```
1. Start
2. Define function f(x)
3. Read lower limit of integration, upper limit of  integration
and number of sub interval
4. Calcultae: step size = (upper limit - lower limit)/number of
sub interval
5. Set: integration value = f(lower limit) + f(upper limit)
6. Set: i = 1
7. If i > number of sub interval then goto
8. Calculate: k = lower limit + i * h
9. If i mod 3 =0 then
     Integration value = Integration Value + 2* f(k)
  Otherwise
    Integration Value = Integration Value + 3 * f(k)
  End If
10. Increment i by 1 i.e. i = i+1 and go to step 7
11. Calculate: Integration value = Integration value * step
size*3/8
12. Display Integration value as required answer
13. Stop
```

OUTPUT:

```
Select B:\3rd Sem\Lab\NM\Lab-3\Integration using Simpsons 3_8 rule.exe
Function is f(x) (pow(x,2)+(sin(x)/x))
Enter lower limit of integration: 2
Enter upper limit of integration: 3
Enter number of sub intervals: 0.167

Required value of integration is: 1.#IO
                This program written by Aaryan Kushawaha ( 28900 )
--------------------------------
```

TASK 4 : Integration using Trapeziodal rule

Algorithm for trapezoidal rule

1. *Start*
2. *Read the value of lower limit of integration (a), upper limit of integration (b) and number of segments (n).*
3. *Compute* $h = (b - a)/n$

   $x = a$
4. *for i=0 to n*

   $y_i = f(x)$

   $x = x + h$

   Repeat i
5. *compute sum=* $y_0 + y_n$
6. *for i=1 to n-1*

   *sum+=2** y_i

   Repeat i
7. *Compute sum=(h/2)* sum*
8. *Display sum as integral value.*
9. *END*

OUTPUT:

B:\3rd Sem\Lab\NM\Lab-3\Integration using Trapeziodal rule.exe

```
Function is f(x) sqrt(sin(x))
Enter lower limit of integration: 1
Enter upper limit of integration: 2
Enter number of sub intervals: 1

Required value of integration is: 0.624
                This program written by Aaryan Kushawaha ( 28900 )
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```

CONCLUSION:

➔Program to implement solving the Derivative using forward difference , integration Simpson 1/3 , 3/8 & Integration using trapeziodal Rule using C Language , were written using High level language & Respective output of each program . Therefore , all results were satisfied by the program.